



Official telegram channel !

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- True or False: Write True if the statement is correct or False if it is incorrect on the space provided (1 pint each)

- _____ 1. The work done by a conservative force along a closed path is always non-zero.
- _____ 2. Both momentum and kinetic energy are conserved in perfectly- inelastic collision.
- _____ 3. Mechanical energy may not be conserved in an isolated spring mass system.
- _____ 4. Uniformly accelerated motion is possible only on a straight path.

Part II - Multiple Choice: Choose the best answer and write the letter of your choice on the space provided (2 pints each)

- _____ 1. One of the following is not true according to Newton's Laws of motion?
a) to every action there is equal and opposite reaction. b) The body is said to be in equilibrium if it is either at rest or in uniform motion.
c) Freely falling object is in gravitational equilibrium
d) none
- _____ 2. Which of the following is not true about uniform circular motion?
a) its acceleration is always radial b) its tangential speed is constant c) It has uniform velocity d) none
e) none
- _____ 3. Object A is fired horizontally with a very large speed while a second object B is released from the same elevation at the same time. Which one of the following is not true about the motion of objects?
a) Object B reaches the ground first b) Object A hits the ground with larger speed
c) Both of them reach the ground equally d) Object A covers larger horizontal range d) none
- _____ 4. Which one of the following motion has non-uniform acceleration?
a) Projectile motion b) uniform circular motion c) uniform motion d) none
- _____ 5. For a given projectile motion, the maximum possible range occurs for the launching angle of
a) 45° b) 60° c) 30° d) 72° e) none
- _____ 6. If a constant force of 100 N is applied for 2 minutes to a 20 kg mass which is initially at rest, the speed acquired and the distance traveled respectively are:
a) 60 m/s and 3.6 km b) 600 m/sec and 36 km c) 600 m/s and 72 km d) none

Part III- Short Answer: Give short and brief explanation for the following questions.

1. State and explain the work energy theorem

(2 pts)

2. What is the center of mass of the system?

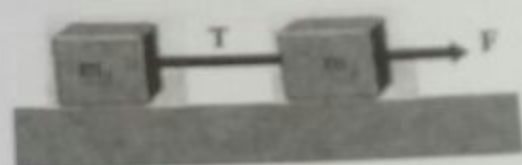
(2 pts)

Part III- Short Answer: Show all the necessary steps with clear hand writing

1. Upstretched rope of negligible mass connects two blocks of masses $m_1 = 12 \text{ kg}$, $m_2 = 18 \text{ kg}$ that are dragged by force $F = 60 \text{ N}$ along a smooth horizontal surface as shown below. Calculate:

- a) the acceleration of the two masses

(3pts)



- b) the tension T in the rope

(3pts)

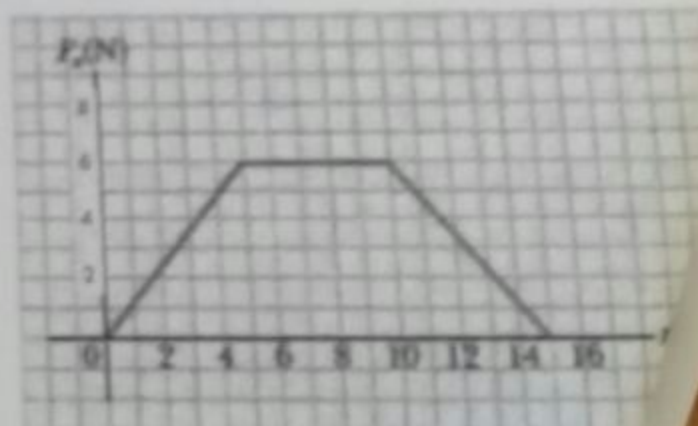
2. A particle of mass 6 kg is subject to a force that varies with position as shown in figure below. Find the work done by the force on the particle as it moves

- a) From $x = 0$ to $x = 5 \text{ m}$

(2pts)

- b) From $x = 5$ to $x = 10 \text{ m}$

(2pts)



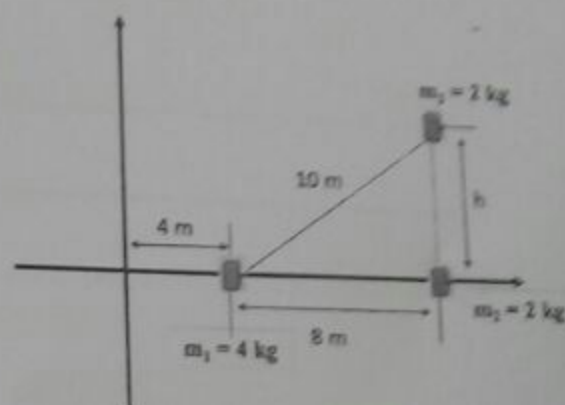
c) From $x = 10 \text{ m}$ to $x = 15 \text{ m}$

(2pts)

d) Find the speed of the particle at 5 m

(4pts)

3. Three particles are located at corner of a right angle triangle as shown in the figure below. Find the center of mass of the system (6pts)



4. A block of mass 680 kg attached to horizontal spring whose spring constant is 65 N/m . the block pulled to a distance of 11 cm from the mean position (Amplitude) and released from rest. Calculate :
- i. Angular frequency (2pts)

ii. Time period (2pts)

iii. Maximum speed (2pts)

iv. Total mechanical energy of SHM system (2pts)